

Replication of Counterfactual Simulatability of Natural Language Explanations

Replication Summary of Chen et al. (ICML 2024)

Abstract

Chen et al. (ICML 2024) study the counterfactual simulatability of natural language explanations generated by large language models. We report a controlled partial replication on StrategyQA, reproducing the Chain-of-Thought (CoT) and Post-Hoc explanation experiments and evaluating simulation precision and counterfactual diversity. Using GPT-OSS 20B and 120B as proxies for GPT-3.5 and GPT-4 with an LLM-based simulator, we obtain 79.57% overall simulation precision, compared with 79.80% in the original paper. SHP and the original human-simulation pipeline were outside the replication scope; this is not a full replication.

1. Experimental Setup

We replicate the StrategyQA component of Chen et al. (2024), “Do Models Explain Themselves? Counterfactual Simulatability of Natural Language Explanations.” For each model explanation, counterfactual questions are generated and a simulator predicts the model’s answer from the explanation. *Simulation precision* is the fraction of simulatable counterfactuals on which the simulated answer matches the model’s actual answer; *counterfactual diversity* is one minus the mean pairwise similarity (BLEU, cosine, Jaccard) between counterfactuals. We replicate Tables 3 (diversity), 4 (precision), 5 (precision–generality correlation) and 6 (task accuracy vs. precision) of the original paper; Tables 3–6 below carry the original numbering, and Tables A–C are our own.

Table A. Comparison of the original experimental setup and our replication.

Component	Original Paper	Our Replication
Dataset	StrategyQA + SHP	StrategyQA
Models	GPT-3.5, GPT-4	GPT-OSS 20B, GPT-OSS 120B
Model correspondence	—	GPT-OSS 20B → proxy for GPT-3.5 GPT-OSS 120B → proxy for GPT-4
Explanation methods	CoT, Post-Hoc	CoT, Post-Hoc
Counterfactual generation	10 for StrategyQA; 6 for SHP	10 per explanation
Simulation	Human simulation	LLM-based simulator
StrategyQA experiment size	—	30 questions × 10 counterfactuals × 4 conditions = 1,200 evaluations

2. Evaluation Workflow

Table C lists which model performs each step of the pipeline in the original study and in our replication.

Table C. Who does what in each step, original paper vs. our replication.

Step	Original paper	Our replication
1. Generate explanations	GPT-3.5 or GPT-4 answers the StrategyQA question and writes a CoT or Post-Hoc explanation.	gpt-oss-20b (small) or gpt-oss-120b (large) answers the question and writes a CoT or Post-Hoc explanation.
2. Generate co- underfactuals	GPT-3.5 and GPT-4 (outputs mixed) write follow-up yes/no questions from the explanation; 10 per explanation.	gpt-oss-120b reads the explanation and writes 10 follow-up yes/no questions whose answers follow from it.
3. Simulate	Three human annotators (Amazon Mechanical Turk) guess the model’s answer from the explanation, or mark it unsimulatable ().	gpt-oss-120b reads the explanation and starter question, and guesses the model’s answer or outputs “cannot guess” ().
4. Get actual answers	The evaluated model (GPT-3.5 or GPT-4) answers each counterfactual.	The evaluated model (gpt-oss-20b or gpt-oss-120b) answers each counterfactual independently.
5. Score	Precision: share of simulatable counterfactuals where guess = actual answer. Generality: $1 - \text{mean pairwise similarity}$.	Same metrics; counterfactuals marked are excluded from precision.

3. Replication Scope and Limitations

Table B. Replication scope and methodological deviations.

Component	Status	Component	Status
StrategyQA	Replicated	Counterfactual diversity	Replicated
CoT / Post-Hoc	Replicated	Precision–generality correlation	Replicated
Simulation precision	Replicated	GPT-3.5/GPT-4 models	Replaced by GPT-OSS proxies

This work should be interpreted as a controlled partial replication. SHP was outside the replication scope. Human simulation was replaced by an LLM-based simulator to obtain a scalable and reproducible evaluation; consequently, the human-vs-LLM validation performed in the original study was not reproduced. In addition, gpt-oss-120b generates the counterfactuals and performs the simulation in all conditions, and in the large condition it also generates the explanations, so the simulator is not independent of the explaining model there. This is a methodological deviation from the human simulation used in the original study.

Result Comparison: Original Paper vs. Replication

Chen et al. (ICML 2024) vs. our controlled partial replication on StrategyQA

4. Main Results

Table 4 reports simulation precision for the four explanation systems. The overall value is the mean across the four conditions.

Original	Replication	Difference
79.80%	79.57%	−0.23 pp
Original → Replication (%): 77.3 → 74.94; 76.8 → 85.79; 81.1 → 85.78; 83.9 → 71.75		

Table 4. Simulation precision (%) across explanation systems. Replication of Table 4.

Model Scale	Explanation	Original	Replication	Δ (pp)
Small	CoT	77.3	74.94	−2.36
Small	Post-Hoc	76.8	85.79	+8.99
Large	CoT	81.1	85.78	+4.68
Large	Post-Hoc	83.9	71.75	−12.15
Overall	—	79.80	79.57	−0.23

Small: GPT-3.5 (original) / GPT-OSS 20B (ours); Large: GPT-4 / GPT-OSS 120B. Δ = replication − original, in percentage points (pp). Original overall value as reported in Table 6 of Chen et al.; the unweighted mean of the four original cells is 79.78.

The aggregate simulation-precision result is closely reproduced (79.80% vs. 79.57%), although individual conditions show variation.

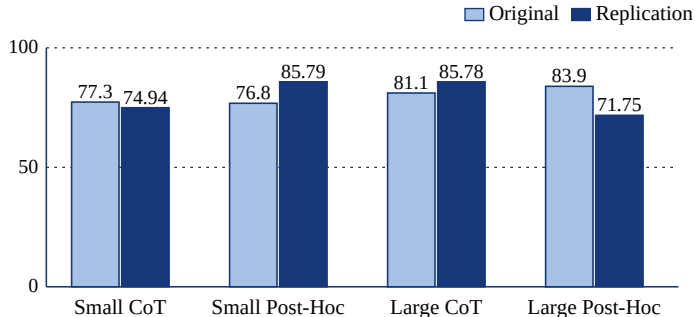


Figure 1. Simulation precision across the four replicated conditions.

5. Additional Results

Table 3. Counterfactual-generation/diversity analysis. Replication of Table 3.

Metric	Original	Replication	Observation
BLEU diversity	67.0–72.9%	87.25%	Higher diversity in replication
Cosine diversity	24.6–29.6%	33.38%	Higher diversity in replication
Jaccard diversity	58.9–66.2%	81.93%	Higher diversity in replication

Original ranges span the GPT-3, GPT-4 and GPT-mix counterfactual generators on StrategyQA. Replication values are averaged across the four conditions.

Table 5. Precision–generality correlation. Replication of Table 5.

Metric	Original	Replication	Observation
Spearman ρ , precision vs. generality	0.017 / 0.002 / −0.007 (BLEU / Cosine / Jaccard)	≈ 0.000	Both near zero

Table 6. Task accuracy and simulation precision. Replication of Table 6.

Metric	Original	Replication	Observation
Average simulation precision	79.80%	79.57%	Near-exact aggregate reproduction
Task accuracy / adherence	75.9% StrategyQA accuracy	100% CoT format / adherence	Different measurement; not directly equivalent

6. Conclusion

Our replication reproduces the central StrategyQA counterfactual-simulatability evaluation and obtains an overall simulation precision of 79.57%, closely matching the original 79.80% result (−0.23 percentage points). The replicated CoT/Post-Hoc and counterfactual-diversity analyses provide additional evidence of consistency with the original evaluation, while differences in model family and simulator implementation limit claims of exact reproduction.